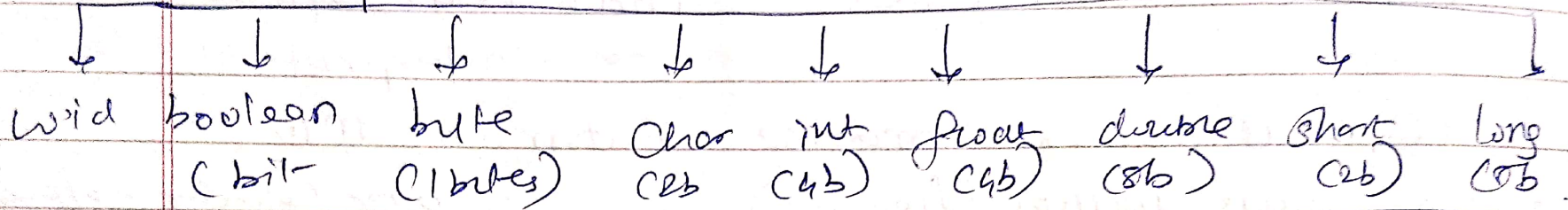


Assignment - 24

1) Datatypes in Java

Primitive



Non-Primitive

Array String Class Interface

2) Why is size of all datatype in Java same on all platform & architecture?

→ Is platform independent & Architecture Neutral

5] Use of UNICODE System as compared to ASCII

ASCII		UNICODE	
Size	1 or 8 bit	16 bit, 32 bit, variable	
Characters	128 or 256 (chars)		
Language support	English only	Supports all language multilingual (Hindi, Arabic, Chinese, etc.)	
Used in	Old systems, old apps	Modern web, Java, XML, databases	

6] What is meant by managed memory?

- - Java manages memory allocation using Garbage Collection (GC).
- Allocates & deallocates memory dynamically preventing memory leaks.
- No explicit memory deallocation required (as free).

7] What are diff types of characteristics (behaviors) in Java?

→ Characteristics:

- 1) Object oriented concepts.
- 2) Platform independent.
- 3) Robust & secure.
- 4) Multithreading.
- 5) Dynamic & distributed.

Behaviors:

- 1) Exhibits behaviors through methods.

8] Prototype of main() in Java.

```
public static void main (String [] args) {}
```

public - Accessible from anywhere.

Static - Can be called w/o creating an object.

void - Does not return any value.

main - Execution starts from here.

String [] args - Command-line arguments.

47 Types of loops in Java

- 1) For Loop : `for (initialization, condition, increment/decrement)`
- When no. of iterations is known.
- 2) While loop : `while (condition) { }`
- When no. of iterations is unknown.
- 3) do-while loop - `do { } while (condition);`
- Executes at least once.
- 4) For-each loop - `for (type var : array) { }`
- Iterates over arrays / collections.

51 How to create an object in Java

- 1) Using 'new' keyword
- 2) Memory allocated for each obj in Java is heap memory.
- If Java file contains multiple classes then, we have to write such a class at top, which contains `main()` init.
- 3) If sequence is not followed, compiler will show error
→ `main()` not found in 1st class.
- 4) For every class in Java, separate class file is created.
- 5) If there are multiple classes in our file, then for each class Java compiler creates diff. class files.

47 Type of Array in Java

- 1) Single-dimensional Array
- 2) 2D / multidimensional Array
- 3) Jagged Array (Diff row sizes)
`int [][] jaggedArr = new int[2][3];`
`jaggedArr[0] = new int[3];`
`jaggedArr[1] = new int[2];`
- 4) Anonymous Array:
`new int[] {1, 2, 3, 4}`

Q] What is meant by object reference in Java?

→ A variable that stores the memory address of an object.

eg. `Car car1 = new Car();` // stores ref. to car object

`Car car2 = car1` // now references the same object

- If we modify `car2`, it also affects `car1` bcoz they point to same object.